

## Discover & Define

### 01 The problem

During peak hours, students usually spend a lot of time finding available seats in Qingfen canteen. However, most of the seats are occupied by bags, which makes it hard for students to tell whether the seat is truly “available” or not.

As a result, the canteen is always over-crowded during peak hours. Students who are in a hurry often feel frustrated and anxious.



Figures 1.1–1.4: Canteen conditions. Source: Photograph by author/team member.

### POV statement

Students in the canteen need a way to quickly and confidently know which seats are actually available because bags placed on seats often create confusion and make them waste time searching during busy lunch hours.

### 02 The analysis

The main reason lies not in the capacity of the canteen but in the overall management and information display. Students don't really know where the available seats are and when the occupied seat will be available. Putting bags on the seats can be considered as both a sign implying the seat is occupied and a simple action to place stuffs.

## Develop & Deliver

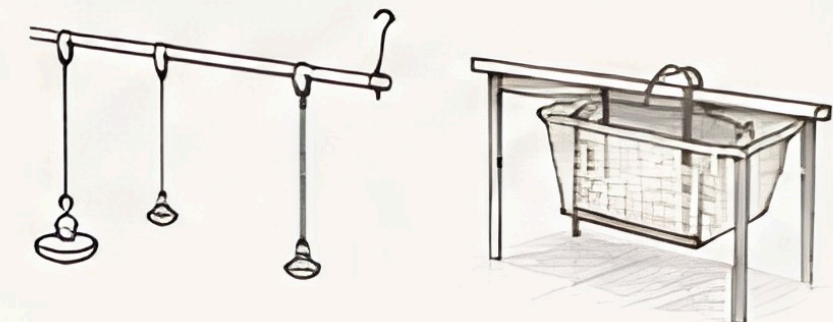
#### 1 OUR INITIAL ASSUMPTION

The issue is bag storage.

Observation



Idea Sketch

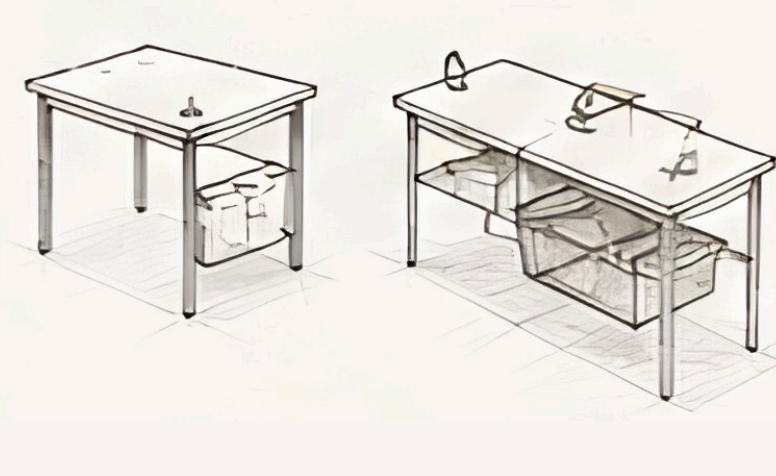


Add hooks or baskets to provide more places for bags.

#### 2 FIRST SOLUTION

We tried storage solutions.

Prototype



Test & Feedback

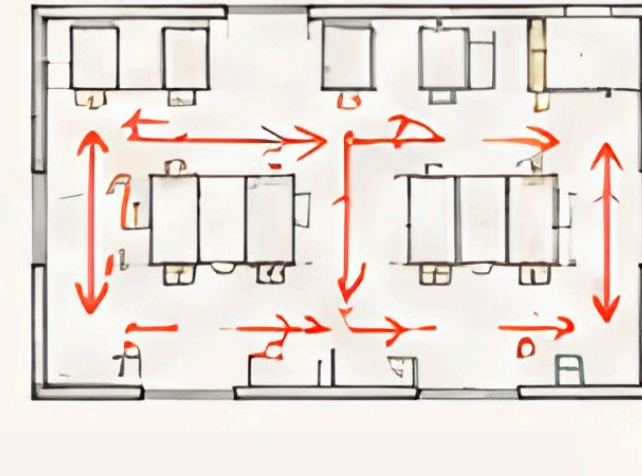


Even with storage, seats were still occupied. The problem remains.

#### 3 DEEPER OBSERVATION

We started to look at how people move.

Flow Study



Key Finding



Students don't know which seats are available. Temporary absence looks the same as vacancy.

#### 4 PROBLEM REFRAMED

The real issue is visibility and uncertainty.

Insight

Lack of visibility

Uncertainty of temporary absence

Inefficient flow & layout

We need a system that makes seat availability visible and trustworthy.

#### 5 SYSTEM DESIGN

Design the system from information to experience.

System Components

Dynamic Monitoring  
Detects movement or presence at the seat.

Seat Status Indicator  
Shows real-time availability status.

Diverse Table Types  
Solo to group seating with flexible configurations.

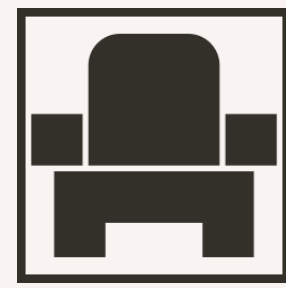
### 03 The final solution

Instead of relying on bags as informal seat markers, the system uses real-time monitoring to detect whether seats are truly occupied. A seat status indicator helps students quickly understand which seats are available, occupied, or temporarily unattended. Combined with diverse table types and a more flexible layout, the system improves seat utilization, reduces unnecessary searching, and creates a calmer dining experience for students during peak hours.



Figure 2: 3D Blender Model. Source: Created by Team Member

### 04 Expected Outcome



#### More Effective Utilization

More seats available when truly free.



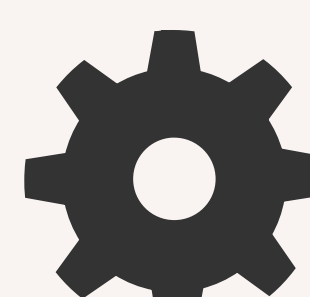
#### Better Experience

Students feel informed, respected, and at ease.



#### Less Time Searching

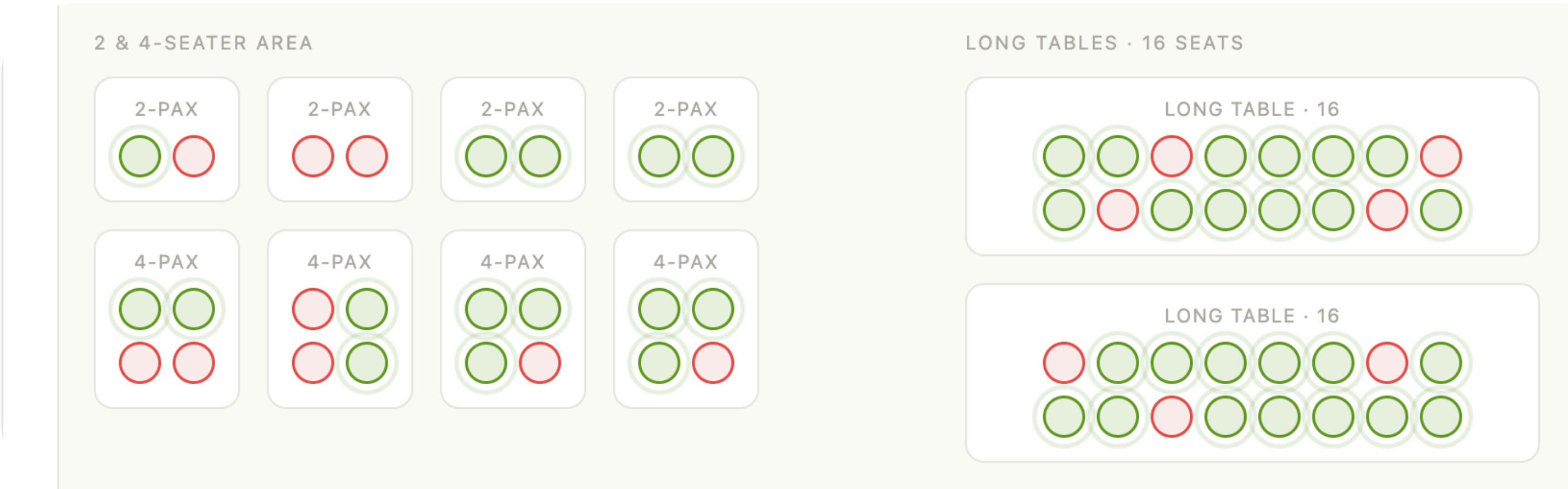
Average seat-hunting time significantly reduced.



#### Systemic Design

Make the whole canteen a interrelated system

Figure 3: Map Display Prototype. Source: Created by Team Member.



## Design Ethics

### 05 Limitations

#### Our Previous Solution

1. Weight System. A system that will detect a person by (weight > 30) = occupied.
2. Alarm for items weighing < 30 kg.

#### Further Judgement

From Design Ethics perspective we notice that the idea of having an alarm might make people uncomfortable, so we decided to go with our current solution, the movement tracker.

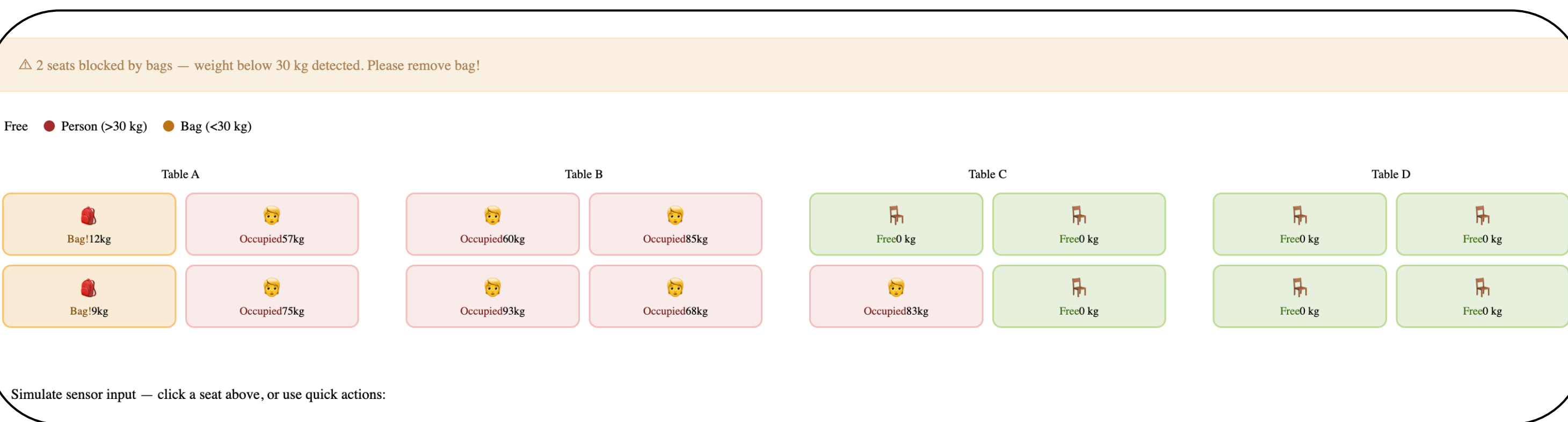


Figure 4: Weight Map Display Prototype. Source: Created by Team Member.